

1. Arctic Cooling Inc. has been in merger talks with Expert HVAC Ltd. Currently both firms are financed with 100% equity. Arctic Cooling, the acquirer, has a market cap of 600 million dollars and 60 million shares outstanding, whereas Expert HVAC has a market cap of 200 million dollars and 50 million shares outstanding.

The management of Arctic Cooling is optimistic toward the synergies between the two companies and estimates the total merger gain would be \$200 million.

The two companies start the negotiation process and Expert HVAC Ltd tells the acquirer the acceptable exchange ratio should be 0.8, which means the acquirer needs to pay 0.8 share to buy 1 share of the target. Assume that the offer price will be paid with 100% stock consideration and that the target shareholders will keep the shares received in the stock payment.

A) What is the number of shares outstanding of the acquirer after the M&A deal?

A = Acquirer; B = Target.

Step #1: Calculate how many shares of A will be paid to B's shareholders

Exchange rate = 0.8: A pays 0.8 new share to buy 1 share of B.

B in total has 50 million shares before the merger.

Therefore # of shares of A paid to purchase B = $0.8 \times 50 \text{ million} = 40 \text{ million shares}$

Step #2: Calculate the number of shares outstanding after the merger

Before merger: A has 60 million shares.

To acquirer B, A pays 40 million new shares.

Total # of shares after merger = $60 \text{ million} + 40 \text{ million} = 100 \text{ million}$.

B) What is the equilibrium share price of the acquirer after the M&A deal?

Total equity value of the merged company = A's standalone equity value + B's standalone equity value + Merger gain = $\$600 + \$200 + \$200 = \$1,000 \text{ million}$.

$$\begin{aligned} \text{Equilibrium share price after merger} &= \frac{\text{Total value of the combined company}}{\text{\# of shares of the combined company}} \\ &= \frac{\$1,000 \text{ million}}{100 \text{ million}} = \$10 \text{ per share} \end{aligned}$$

C) What is the NPV of the merger deal from the target shareholders' perspective?

What the target shareholder receive: 40 million shares in the combined company (see step 1).

Total value of the offer = $40 \text{ million} \times \$10 \text{ from part B)} = \400 million .

What the target surrenders: the target's pre-merger total value = \$200 million.

Net gain or NPV from the target's perspective = \$400 - \$200 = \$200 million.

D) If the exchange ratio is 0.8, what is the NPV from the acquirer shareholders's perspective? Assume that the target shareholders will keep the shares received in stock payment.

The total NPV generated by this M&A deal = \$200 million.

The target's piece of NPV = \$200 million from part C).

The acquirer's piece of NPV = \$200 - \$200 = \$0.

2. Middle Crust Resources Ltd. has been in merger talks with GCB group. Currently both firms are financed with 100% equity. Middle Crust Resources, the acquirer, has a market cap of 600 million dollars, whereas GCG Group has a market cap of 200 million dollars. Both firms have 100 million shares outstanding.

The management of the acquirer estimates that the total merger gain would be \$200 million.

If the exchange ratio is 0.6, what is the NPV of the merger deal from the acquirer shareholders' perspective? Assume that the offer price will be paid with 100% stock consideration and that the target shareholders will keep the shares received in the stock payment.

A = Acquirer; B = Target.

Step #1: Calculate how many shares of A will be paid to B's shareholders

of shares of A needs to be paid = exchange ratio * target's # of shares outstanding = 0.6 * 100 million = 60 million

Step #2: Calculate the number of shares outstanding of the combined company after the merger

A's # of shares outstanding before merger = 100 million

of shares of A to be paid to buy B = 60 million

Total # of shares of the combined company post-merger = 100 + 60 = 160 million

Step #3: Calculate the equilibrium share price after merger

Total equity value of the merged company = A's standalone equity value + B's standalone equity value + Merger gain = \$600 + \$200 + \$200 = \$1,000 million.

Equilibrium share price after merger = $\frac{\text{Total value of the combined company}}{\text{\# of shares of the combined company}} = \frac{\$1,000 \text{ million}}{160 \text{ million}} = \6.25

Step #4: Calculate the NPV to the target

What the target receives: 60 million shares in the combined company (see step 1).

Total value of the offer received by target = 60 million * \$6.25 = \$375 million.

What the target surrenders: the target's pre-merger total value = \$200 million.

Net gain or NPV from the target's perspective = \$375 - \$200 = \$175 million.

Step #5: Calculate the NPV to the acquirer

The total NPV generated by this M&A deal = \$200 million.

The target's piece of NPV = \$175 million.

The acquirer's piece of NPV = \$200 - \$175 = \$25 million.

3. Frost Systems Ltd. is negotiating to acquire CoolBreeze Technologies Inc. Both are 100% equity-financed. Frost has a market cap of 800 million dollars with 80 million shares; CoolBreeze has a market cap of 400 million dollars with 40 million shares.

The estimated merger gain is 250 million dollars. CoolBreeze proposes an exchange ratio of 1.5 (1.5 Frost share for each CoolBreeze share).

What will Frost's total shares be outstanding after the merger?

A = Acquirer; B = Target.

Step #1: Calculate how many shares of A will be paid to B's shareholders

of shares of A needs to be paid = exchange ratio * target's # of shares outstanding = 1.5 * 40 million = 60 million

Step #2: Calculate the number of shares outstanding of the combined company after the merger

A's # of shares outstanding before merger = 80 million

of shares of A to be paid to buy B = 60 million

Total # of shares of the combined company post-merger = 80 + 60 = 140 million

4. GlacierTech Ltd. is negotiating to acquire ArcticFlow Inc. Both are 100% equity-financed. GlacierTech has a market cap of 900 million dollars with 90 million shares, and ArcticFlow has a market cap of 300 million dollars with 30 million shares.

The estimated merger gain is 200 million dollars. ArcticFlow proposes an exchange ratio of 1.2 (1.2 GlacierTech shares for each ArcticFlow share). Assume that the offer price is paid with 100% stock consideration and that the target shareholders will keep the shares received in the stock payment. What will be GlacierTech's equilibrium share price after the merger?

A = Acquirer; B = Target.

Step #1: Calculate how many shares of A will be paid to B's shareholders

of shares of A needs to be paid = exchange ratio * target's # of shares outstanding = $1.2 * 30$ million = 36 million

Step #2: Calculate the number of shares outstanding of the combined company after the merger

A's # of shares outstanding before merger = 90 million

of shares of A to be paid to buy B = 36 million

Total # of shares of the combined company post-merger = $90 + 36 = 126$ million

Step #3: Calculate the equilibrium share price after merger

Total equity value of the merged company = A's standalone equity value + B's standalone equity value + Merger gain = $\$900 + \$300 + \$200 = \$1,400$ million.

Equilibrium share price after merger = $\frac{\text{Total value of the combined company}}{\text{\# of shares of the combined company}} = \frac{\$1400 \text{ million}}{126 \text{ million}} = \11.11

5. Summit Cooling Ltd. is negotiating to acquire BreezeAir Technologies Inc. Both companies are 100% equity-financed. Summit Cooling has a market cap of 700 million dollars with 70 million shares, while BreezeAir has a market cap of 150 million dollars with 15 million shares.

The proposed offer price is 20 dollars per share, with 40% paid in cash and 60% paid in stock. The cash consideration will be financed with debt. The acquirer can issue new shares at its market value. What should the exchange ratio be if the target accepts the offer price?

A = Acquirer; B = Target.

Step #1: Calculate stock consideration per share

Stock consideration per share = $60\% * \$20 = \12 per share

Cash consideration per share = $40\% * \$20 = \8 per share

Step #2: Calculate the acquirer's new share issuance price

The acquirer's new share issuance price = the acquirer's market share price * (1- issuance discount). In this case, the acquirer can issue new shares at the market value. So, the new share issuance price = the market value = $700/70 = \$10$

Step #3: Calculate the exchange ratio

Note that the exchange ratio is calculated as **stock consideration per share divided by the acquirer new share issuance price (NOT the offer price divided by the acquirer new share issuance price)**.

$$\text{Exchange ratio} = \$12/\$10 = 1.2$$

It means that each share of the target will receive a combination of 1.2 acquirer shares plus \$8 of cash.

6. PolarTech Ltd. is negotiating to acquire NorthBreeze Solutions Inc. Both companies are 100% equity-financed. PolarTech has a market capitalization of 800 million dollars with 80 million shares, while NorthBreeze has a market capitalization of 200 million dollars with 20 million shares.

The proposed offer price is 18 dollars per share, with 50% paid in cash and 50% paid in stock. The cash consideration will be financed with debt. The acquirer can issue new shares at its market value. What should the number of shares outstanding of the acquirer after the M&A if the target accepts the offer price?

A = Acquirer; B = Target.

Step #1: Calculate stock consideration per share

$$\text{Stock consideration per share} = 50\% * \$18 = \$9 \text{ per share}$$

Step #2: Calculate the acquirer's new share issuance price

The acquirer's new share issuance price = the acquirer's market share price * (1- issuance discount). In this case, the acquirer can issue new shares at the market value. So, the new share issuance price = the market value = $200/20 = \$10$

Step #3: Calculate the exchange ratio

Note that the exchange ratio is calculated as stock consideration per share divided by the acquirer new share issuance price (NOT the offer price divided by the acquirer new share issuance price).

$$\text{Exchange ratio} = \$9/\$10 = 0.9$$

Step #4: Calculate how many shares of A will be paid to B's shareholders

$$\# \text{ of shares of A needs to be paid} = \text{exchange ratio} * \text{target's \# of shares outstanding} = 0.9 * 20 \text{ million} = 18 \text{ million}$$

Step #5: Calculate the number of shares outstanding of the combined company after the merger

A's # of shares outstanding before merger = 80 million

of shares of A to be paid to buy B = 18 million

Total # of shares of the combined company post-merger = $80 + 18 = 98$ million

7. Alpha Ltd. acquires 100% of Beta Ltd.'s equity for a total purchase price of \$50 million, fully paid in cash. At the time of acquisition, Beta Ltd.'s book value of total assets is \$40 million, and the book value of total liabilities is \$20 million. However, the fair value of the target's total assets is adjusted upward by \$10 million, with no adjustment to liabilities.

What is the amount of goodwill recorded from the acquisition?

Step 1: Fair value of total assets of the target

Fair value of total assets of the target = $\$40 + \text{adjustment} = \$40 + \$10 = \50 million

Step 2: Fair value of total liabilities of the target

Fair value of total liabilities of the target = $\$20 + \text{adjustment} = \$20 + \$0 = \20 million

Step 3: Fair net value of assets of the target

Fair net value of assets of the target = $\$50 - \$20 = \$30$ million

Step 4: Goodwill

Total offer price – fair net value of assets = $\$50 - \$30 = \$20$ million

8. Vega Ltd. acquires 100% of Orion Ltd.'s equity for a total purchase price of \$60 million, fully paid in cash. At the time of acquisition, Orion Ltd.'s book value of total assets is \$85 million, and the book value of total liabilities is \$40 million. The fair value of Orion's total liabilities is adjusted upward by \$10 million, with no adjustment to assets.

What is the amount of goodwill recorded from the acquisition?

Step 1: Fair value of total assets of the target

Fair value of total assets of the target = $\$85 + \text{adjustment} = \$85 + \$0 = \85 million

Step 2: Fair value of total liabilities of the target

Fair value of total liabilities of the target = $\$40 + \text{adjustment} = \$40 + \$10 = \50 million

Step 3: Fair net value of assets of the target

Fair net value of assets of the target = $\$85 - \$50 = \$35$ million

Step 4: Goodwill

Total offer price – fair net value of assets = $\$60 - \$35 = \$25$ million

9. Summit Cooling Ltd. plans to acquire 100% of BreezeAir Technologies Inc.'s equity using high leverage. The interest expenses for the debt financing of this LBO transaction are expected to be \$5 million, \$4 million, and \$4 million in years 1, 2, and 3 after the acquisition, respectively. The acquirer's tax rate is 30%, and the cost of debt is 6%.

What is the present value (PV) of these interest tax shields over the three years?

Step 1: Interest tax shield

$$\text{Year 1 : interest exp} * \text{tax rate} = \$5 * 0.3 = \$1.5$$

$$\text{Year 2 : interest exp} * \text{tax rate} = \$4 * 0.3 = \$1.2$$

$$\text{Year 3 : interest exp} * \text{tax rate} = \$4 * 0.3 = \$1.2$$

Step 2: PV of interest tax shield

Should use the cost of debt as the discount rate since the interest tax shield is part of cash flow from its debt.

$$PV = \frac{1.5}{(1+0.06)} + \frac{1.2}{(1+0.06)^2} + \frac{1.2}{(1+0.06)^3} = \$3.49 \text{ million}$$

10. GlacierTech Ltd. plans to acquire 100% of Polar Systems Inc.'s equity through a leveraged buyout (LBO). The interest expenses from the debt financing are projected to be \$6 million, \$5 million, and \$4 million in years 1, 2, and 3 after the acquisition, respectively. GlacierTech's tax rate is 30%, and its cost of debt is 7%.

What is the present value (PV) of these interest tax shields over the three years?

Step 1: Interest tax shield

$$\text{Year 1 : interest exp} * \text{tax rate} = \$6 * 0.3 = \$1.8$$

$$\text{Year 2 : interest exp} * \text{tax rate} = \$5 * 0.3 = \$1.5$$

$$\text{Year 3 : interest exp} * \text{tax rate} = \$4 * 0.3 = \$1.2$$

Step 2: PV of interest tax shield

- Should use the cost of debt as the discount rate since the interest tax shield is part of cash flow from its debt.

- $$PV = \frac{1.8}{(1+0.07)} + \frac{1.5}{(1+0.07)^2} + \frac{1.2}{(1+0.07)^3} = \$3.97 \text{ million}$$